Randy Pahang II

+1 (xxx) xxx-xxxx | example@gmail.com | linkedin.com/in/randypahangii | github.com/randyp2 | randypahangii.com

Education

University of Nevada, Las Vegas Expected May 2027
B.S. Computer Science | *Honors College* | *Wolzinger Family Engineering Scholar* *GPA: 4.0/4.0*

Experience

JT4 (Amentum) May 2026 – Present
Software Engineering Intern *Las Vegas, NV*
- Built a C++23 telemetry simulator sustaining 20 Mbps of real-time flight-test data with precise packet timing, replacing commercial software and saving **$10K+** in licensing costs.
- Designed a configurable packet-generation pipeline reproducing flight-representative timing and data rates to validate downstream telemetry processing before flight-test and range operations.
- Exposed the C++ engine via a C ABI to a Python/PySide6 dashboard showing 100+ live telemetry channels.
- Built a GoogleTest suite with **93% coverage** and validated output against recorded flight-test data.

STARS Solution LLC Nov 2025 – Jun 2026
Software Engineering Intern (Part-Time) *Las Vegas, NV*
- Built Java Spring Boot REST APIs integrating data from 30,000+ schools for a 3,500+ user pilot.
- Cut LLM recommendation time-to-first-response from **8.2s to 0.4s** by moving generation to asynchronous RabbitMQ workers and streaming partial results via Server-Sent Events.
- Cut institution-search latency from **110ms to 37ms** with a Redis cache-aside layer and versioned invalidation.
- Built production monitoring and alerting for AI workloads with Micrometer, AWS CloudWatch, and SNS, catching LLM provider failures, token exhaustion, and failed jobs.

Leadership

Bit2Byte Nov 2025 – Present
Co-Founder & Lead Developer *Las Vegas, NV*
- Co-founded UNLV's software development organization (100+ members), running Agile sprint planning, code reviews, and workshops for student teams building with React/Next.js, Spring Boot, and FastAPI.

Projects

PortRN | *Next.js, TypeScript, Spring Boot, PostgreSQL, RabbitMQ, Docker, AWS ECS* Nov 2025 – Apr 2026
- Built and launched a full-stack AI portfolio and networking platform with a Next.js/React frontend and Spring Boot API, containerized with Docker on AWS ECS for a 50+ user beta, with idempotent Stripe billing.
- Parallelized multi-step LLM generation across asynchronous RabbitMQ workers with section-level validation and repair of model output, cutting end-to-end processing from **7–10 to 4–5 minutes**.
- Built a skill-verification engine that checks resume claims against GitHub commit history, discounting forked or copied code so scores reflect each user's own work.
- Replaced unreliable iframe rendering of AI-generated React components by transpiling them in-browser with Babel against a fixed library scope, with cached compiled output.
- Set up GitHub Actions CI that builds both apps and runs **800+ JUnit and Vitest tests** on every push and PR.

1st of 100+ Participants, RebelHacks 2026 | *HotelGuard* Jan 2026
- Built a concurrent pipeline pairing YOLOv8 tracking with parallel LLMs to score room-level risk.

TrafficControl | *Go, TCP/UDP, Linux, YAML* Sep 2026 – Present
- Built a Go network traffic generator that drives concurrent TCP/UDP streams from declarative YAML configs using goroutines and channels, sustaining **150K UDP packets/s**.
- Kept send rates exact under OS scheduling delays with a deadline-based pacer using fractional-nanosecond carry and catch-up bounded to a 5 ms window, reaching 0.0007% median pacing error across five 10s loopback runs.

Technical Skills

Languages: C/C++, Java, Python, Go, TypeScript, JavaScript, SQL
Frameworks & Data: Spring Boot, React, Next.js, FastAPI, Node.js, PostgreSQL, Redis, RabbitMQ
Cloud & DevOps: AWS (ECS, EC2, S3, CloudWatch), Docker, GitHub Actions (CI/CD), Git, Linux
Testing & Build: GoogleTest, JUnit, Vitest, CMake